



Diesel Generator Set

mtu 12V4000 DS1750

380V – 11 kV/50 Hz/prime power/fuel consumption optimized
12V4000G14F/water charge air cooling



Optional equipment and finishing shown. Standard may vary.

Product highlights

Benefits

- Low fuel consumption
- Optimized system integration ability
- High reliability
- High availability of power
- Long maintenance intervals

Support

- Global product support offered

Standards

- Engine-generator set is designed and manufactured in facilities certified to standards ISO 2008:9001 and ISO 2004:14001
- Generator set complies to ISO 8528
- Generator meets NEMA MG1, BS 5000, ISO, DIN EN and IEC standards
- NFPA 110

Power rating

- System ratings: 1590 kVA - 1700 kVA
- Accepts rated load in one step per NFPA 110*
- Generator set complies to G3 according to ISO 8528-5
- Generator set exceeds load steps according to ISO 8528-5*

Performance assurance certification (PAC)

- Engine-generator set tested to ISO 8528-5 for transient response
- 75% load factor
- Verified product design, quality and performance integrity
- All engine systems are prototype and factory tested

Complete range of accessories available

- Control panel
- Power panel
- Circuit breaker/power distribution
- Fuel system
- Fuel connections with shut-off valve mounted to base frame
- Starting/charging system
- Exhaust system
- Mechanical and electrical driven radiators
- Medium and oversized voltage alternators

Emissions

- Fuel consumption optimized

Certifications

- CE certification option
- Unit certificate acc. to VDE-AR-N 4110

* Changes to the standard parameter sets (alternator-regulator and genset-controller) are necessary



A Rolls-Royce
solution

Application data ¹⁾

Engine			Liquid capacity (lubrication)	
Manufacturer		<i>mtu</i>	Total oil system capacity: l	260
Model		12V4000G14	Engine jacket water capacity: l	160
Type		4-cycle	Intercooler coolant capacity: l	40
Arrangement		12V	Combustion air requirements	
Displacement: l		57.2	Combustion air volume: m³/s	1.6
Bore: mm		170	Max. air intake restriction: mbar	50
Stroke: mm		210	Cooling/radiator system	
Compression ratio		16.4	Coolant flow rate (HT circuit): m³/hr	56
Rated speed: rpm		1500	Coolant flow rate (LT circuit): m³/hr	30
Engine governor		ECU 9	Heat rejection to coolant: kW	540
Max power: kWm		1420	Heat radiated to charge air cooling: kW	200
Air cleaner		dry	Heat radiated to ambient: kW	75
Fuel system			Fan power for electr. radiator (40°C): kW	38
Maximum fuel lift: m		5	Exhaust system	
Total fuel flow: l/min		16	Exhaust gas temp. (after turbocharger): °C	460
Fuel consumption ²⁾			Exhaust gas volume: m³/s	4.0
At 100% of power rating:	l/hr	g/kwh	Maximum allowable back pressure: mbar	85
At 75% of power rating:	323.3	189	Minimum allowable back pressure: mbar	30
At 50% of power rating:	250.2	195		
	173.7	203		

Standard and optional features

System ratings (kW/kVA)

Generator model	Voltage	fuel consumption optimized					
		without radiator			with mechanical radiator		
		kWel	kVA*	AMPS	kWel	kVA*	AMPS
Leroy Somer LSA52.3 S5 (Low voltage Leroy Somer standard)	380 V	1360	1700	2583	1320	1650	2507
	400 V	1360	1700	2454	1320	1650	2382
	415 V	1360	1700	2365	1320	1650	2295
Leroy Somer LSA52.3 S6 (Low voltage Leroy Somer oversized)	380 V	1360	1700	2583	1320	1650	2507
	400 V	1360	1700	2454	1320	1650	2382
	415 V	1360	1700	2365	1320	1650	2295
Marathon 743RSL7090 (Low voltage Marathon)	380 V	1352	1690	2568	1312	1640	2492
	400 V	1344	1680	2425	1312	1640	2367
	415 V	1272	1590	2212	1272	1590	2212
Marathon 744RSL7091 (Low voltage Marathon oversized)	380 V	1352	1690	2568	1312	1640	2492
	400 V	1344	1680	2425	1312	1640	2367
	415 V	1272	1590	2212	1272	1590	2212
Marathon 1020FDH7095 (Medium volt. marathon)	11 kV	1352	1690	89	1312	1640	86
Leroy Somer LSA53.2 VL6 (Medium volt. Leroy Somer)	11 kV	1352	1690	89	1320	1650	87

* cos phi = 0.8

1 All data refers only to the engine and is based on ISO standard conditions (25°C and 100m above sea level).
2 Values referenced are in accordance with ISO 3046-1. Conversion calculated with fuel density of 0.83 g/ml. All fuel consumption values refer to rated engine power.